

B6 Condensed-Matter Physics — Model Solutions, 2016

Question 1 — X-ray diffraction, structure factor, compressibility

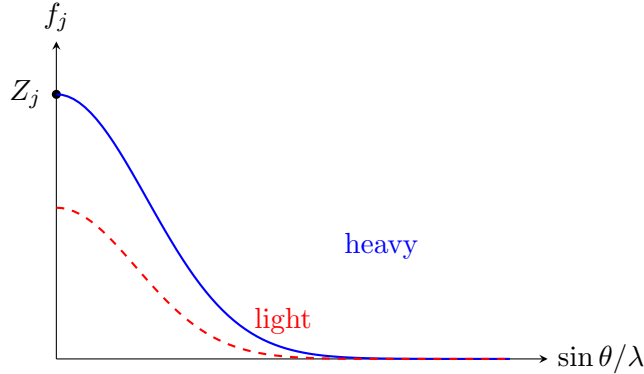
Problem. Define structure factor and atomic form factor; sketch $f_j(\theta)$; relate intensity to $F(hkl)$. Treat CsCl, then Fe (BCC). Derive the pressure-shift formula $(d\theta/dp)_T = \alpha\kappa \tan \theta$ for an Fe sample and compute $2\theta_{110}$ at ambient and at 100 GPa.

(a) Structure factor and form factor

Structure factor for the hkl reflection:

$$F(hkl) = \sum_{j \in \text{basis}} f_j \exp[2\pi i(hx_j + ky_j + lz_j)]. \quad (1)$$

The atomic form factor $f_j(\mathbf{Q})$ is the Fourier transform of the electron density of atom j . Forward scattering: all electrons scatter in phase, so $f_j(\theta = 0) = Z_j$. As θ grows the scatterers within the atom de-phase and f_j falls.



Diffracted intensity:

$$I(hkl) \propto |F(hkl)|^2.$$

(b) CsCl and the Fe analogue

CsCl basis: Cs^+ at $(0, 0, 0)$ with form factor f_{Cs} ; Cl^- at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ with f_{Cl} .

Apply (??):

$$F(hkl) = f_{\text{Cs}} + f_{\text{Cl}} e^{i\pi(h+k+l)}.$$

Use $e^{i\pi n} = (-1)^n$:

$$F(hkl) = f_{\text{Cs}} + (-1)^{h+k+l} f_{\text{Cl}}.$$

Square modulus:

$$I(hkl) \propto [f_{\text{Cs}} + (-1)^{h+k+l} f_{\text{Cl}}]^2 \quad (2)$$

Replace both ions by Fe ($f_{\text{Cs}}, f_{\text{Cl}} \rightarrow f_{\text{Fe}}$):

$$F(hkl) = f_{\text{Fe}} [1 + (-1)^{h+k+l}].$$

Square:

$$I(hkl) \propto f_{\text{Fe}}^2 [1 + (-1)^{h+k+l}]^2 = \begin{cases} 4f_{\text{Fe}}^2 & h+k+l \text{ even} \\ 0 & h+k+l \text{ odd} \end{cases}.$$

For the (111) reflection, $h+k+l = 3$ (odd):

$$I_{\text{CsCl}}(111) \propto (f_{\text{Cs}} - f_{\text{Cl}})^2, \quad I_{\text{Fe}}(111) = 0.$$

The (111) reflection is allowed for CsCl but *systematically absent* for the Fe-substituted crystal.

The Fe structure (simple cubic with identical atoms at 0 and $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) is body-centred cubic (BCC); allowed reflections satisfy $h+k+l$ even.

(c) Pressure dependence of θ

Bragg's law for cubic lattice:

$$\lambda = 2d_{hkl} \sin \theta, \quad d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}. \quad (3)$$

Combine:

$$\sin \theta = \frac{\lambda \sqrt{h^2 + k^2 + l^2}}{2a}.$$

Take logarithmic derivative at fixed λ :

$$\frac{d \sin \theta}{\sin \theta} = -\frac{da}{a}.$$

Use $d \sin \theta = \cos \theta d\theta$:

$$\cot \theta d\theta = -\frac{da}{a}.$$

Volume-lattice relation: $V = a^3$, so $dV/V = 3 da/a$:

$$\frac{da}{a} = \frac{1}{3} \frac{dV}{V}.$$

Definition of isothermal compressibility:

$$\kappa = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T.$$

At fixed T :

$$\frac{1}{a} \left(\frac{\partial a}{\partial p} \right)_T = -\frac{1}{3} \kappa.$$

Substitute into $\cot \theta d\theta = -da/a$:

$$\cot \theta \left(\frac{\partial \theta}{\partial p} \right)_T = \frac{1}{3} \kappa.$$

Rearrange:

$$\boxed{\left(\frac{\partial \theta}{\partial p} \right)_T = \frac{1}{3} \kappa \tan \theta, \quad \alpha = \frac{1}{3}.} \quad (4)$$

(d) Numerical evaluation

Data: $a_0 = 0.287 \text{ nm}$, $\lambda = 0.154 \text{ nm}$, $\kappa = 0.0059 \text{ GPa}^{-1}$, $hkl = 110$ so $\sqrt{h^2 + k^2 + l^2} = \sqrt{2}$.

(i) **Ambient pressure.** Apply (??):

$$\sin \theta_0 = \frac{\lambda \sqrt{2}}{2a_0} = \frac{0.154 \times 1.4142}{2 \times 0.287}.$$

Numerator:

$$0.154 \times 1.4142 = 0.2178.$$

Divide:

$$\sin \theta_0 = \frac{0.2178}{0.574} = 0.3795.$$

Invert:

$$\theta_0 = \arcsin(0.3795) = 22.30^\circ.$$

$$\boxed{2\theta_0 = 44.6^\circ.}$$

(ii) **Pressure** $p = 100 \text{ GPa}$. Lattice contraction (linearised, treating κ constant):

$$\frac{\Delta a}{a_0} = -\frac{1}{3}\kappa \Delta p.$$

Numerical:

$$\frac{\Delta a}{a_0} = -\frac{1}{3} \times 0.0059 \times 100 = -0.1967.$$

New lattice constant:

$$a = a_0(1 + \Delta a/a_0) = 0.287 \times (1 - 0.1967) = 0.2305 \text{ nm}.$$

Apply (??) at the new a :

$$\sin \theta = \frac{\lambda \sqrt{2}}{2a} = \frac{0.2178}{0.4610} = 0.4724.$$

Invert:

$$\theta = \arcsin(0.4724) = 28.20^\circ.$$

$$\boxed{2\theta = 56.4^\circ.}$$

(The linearised approximation $\Delta a/a = -\frac{1}{3}\kappa \Delta p$ over-estimates the contraction at 100 GPa; the answer is reliable to $\sim 1^\circ$.)

Question 2 — Free-electron model in d dimensions

Problem. Define the free-electron model, Fermi energy and Fermi temperature; explain the room-temperature feel of metals. Derive E_F and $g(E_F)$ in $d = 1, 2, 3$, finding a_d, b_d . Estimate E_F, T_F, τ, ℓ for Al and a 2D electron gas.

(a) Definitions and qualitative answer

Free-electron model: valence electrons treated as a gas of non-interacting fermions of mass m moving in a flat potential, confined to volume $V = L^d$ with periodic boundary conditions; lattice ions provide only neutralising background.

Fermi energy E_F : energy of the highest occupied single-particle state at $T = 0$.

Fermi temperature $T_F = E_F/k_B$: the temperature at which $k_B T$ becomes comparable to E_F .

For typical metals $n \sim 10^{29} \text{ m}^{-3}$ gives $E_F \sim 5 \text{ eV}$ and $T_F \sim 10^4 \text{ K} \gg 300 \text{ K}$. Cold-to-touch sensation arises from the *lattice* (phonon) heat capacity and the high electronic thermal conductivity transporting heat from the skin; the electronic heat capacity is $\sim T/T_F \ll 1$ of the classical value, but the phonon contribution at room temperature is the dominant thermal reservoir, and conduction electrons rapidly remove thermal energy from the skin.

(b) Fermi energy and density of states in d dimensions

Allowed $\mathbf{k}$ states have density $V/(2\pi)^d$ with spin-degeneracy $g_s = 2$.

Fill a Fermi sphere of radius k_F :

$$N = 2 \cdot \frac{V}{(2\pi)^d} \cdot \Omega_d k_F^d / d,$$

where $\Omega_d k_F^{d-1}$ is the surface area of a d -sphere of radius k_F ($\Omega_1 = 2$, $\Omega_2 = 2\pi$, $\Omega_3 = 4\pi$); $\Omega_d k_F^d / d$ is its volume.

Solve for k_F^d :

$$k_F^d = \frac{d(2\pi)^d}{2\Omega_d} n.$$

Use $E_F = \hbar^2 k_F^2 / (2m)$, hence $k_F^2 = (k_F^d)^{2/d}$:

$$E_F = \frac{\hbar^2}{2m} \left(\frac{d(2\pi)^d}{2\Omega_d} \right)^{2/d} n^{2/d} = \frac{\hbar^2}{2m} (n a_d)^{2/d}. \quad (5)$$

Read off $a_d = d(2\pi)^d / (2\Omega_d)$:

$d = 1$: $a_1 = 1 \cdot 2\pi / (2 \cdot 2) = \pi/2$.

$d = 2$: $a_2 = 2(2\pi)^2 / (2 \cdot 2\pi) = 2\pi$.

$d = 3$: $a_3 = 3(2\pi)^3 / (2 \cdot 4\pi) = 3\pi^2$.

Density of states per unit volume: count states with $E' < E$, then differentiate.

Use $N(E)/V = (2/\Omega'_d) \cdot (\Omega_d/d) \cdot k(E)^d / (2\pi)^d \cdot 2$ – equivalently, recall that

$$N(E_F) = N \Rightarrow n = \frac{2\Omega_d}{d(2\pi)^d} k_F^d.$$

For arbitrary E :

$$\frac{N(E)}{V} = \frac{2\Omega_d}{d(2\pi)^d} k(E)^d, \quad k(E) = \sqrt{2mE}/\hbar.$$

Differentiate w.r.t. E to obtain $g(E) = d(N/V)/dE$:

$$g(E) = \frac{2\Omega_d}{d(2\pi)^d} d k^{d-1} \frac{dk}{dE} = \frac{2\Omega_d}{(2\pi)^d} k^{d-1} \frac{dk}{dE}.$$

Use $dk/dE = m/(\hbar^2 k)$:

$$g(E) = \frac{2\Omega_d}{(2\pi)^d} \frac{m k^{d-2}}{\hbar^2}.$$

Evaluate at E_F and divide by n :

$$\frac{g(E_F)}{n} = \frac{2\Omega_d}{(2\pi)^d} \frac{m k_F^{d-2}}{\hbar^2} \cdot \frac{d(2\pi)^d}{2\Omega_d k_F^d} = \frac{dm}{\hbar^2 k_F^2} = \frac{d}{2E_F}.$$

Hence

$$\boxed{g(E_F) = \frac{d}{2} \frac{n}{E_F}, \quad b_d = d/2.} \quad (6)$$

So $b_1 = \frac{1}{2}$, $b_2 = 1$, $b_3 = \frac{3}{2}$.

(c) Aluminium

FCC, 4 atoms per conventional cell, valence 3 $\Rightarrow$ 12 electrons per cell of side $a = 0.405$ nm.

Electron density:

$$n = \frac{12}{a^3} = \frac{12}{(0.405 \times 10^{-9})^3}.$$

Cube the lattice constant:

$$a^3 = 6.64 \times 10^{-29} \text{ m}^3.$$

Divide:

$$n = \frac{12}{6.64 \times 10^{-29}} = 1.81 \times 10^{29} \text{ m}^{-3}.$$

Fermi wavevector ($d = 3$, $a_3 = 3\pi^2$):

$$k_F = (3\pi^2 n)^{1/3} = (3\pi^2 \times 1.81 \times 10^{29})^{1/3}.$$

Argument:

$$3\pi^2 \times 1.81 \times 10^{29} = 5.36 \times 10^{30} \text{ m}^{-3}.$$

Cube root:

$$k_F = 1.75 \times 10^{10} \text{ m}^{-1}.$$

Fermi energy:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{(1.055 \times 10^{-34})^2 (1.75 \times 10^{10})^2}{2 \times 9.11 \times 10^{-31}}.$$

Numerator:

$$(1.055 \times 10^{-34})^2 = 1.113 \times 10^{-68}, \quad (1.75 \times 10^{10})^2 = 3.06 \times 10^{20}.$$

Multiply:

$$1.113 \times 10^{-68} \times 3.06 \times 10^{20} = 3.41 \times 10^{-48}.$$

Divide:

$$E_F = \frac{3.41 \times 10^{-48}}{1.822 \times 10^{-30}} = 1.87 \times 10^{-18} \text{ J} = \boxed{11.7 \text{ eV}}.$$

Fermi temperature:

$$T_F = E_F/k_B = 1.87 \times 10^{-18} / 1.381 \times 10^{-23} = \boxed{1.36 \times 10^5 \text{ K}}.$$

Fermi velocity:

$$v_F = \hbar k_F / m_e = 1.055 \times 10^{-34} \times 1.75 \times 10^{10} / 9.11 \times 10^{-31} = 2.03 \times 10^6 \text{ m s}^{-1}.$$

Drude scattering time from $\rho = m_e / (ne^2\tau)$:

$$\tau = \frac{m_e}{ne^2\rho} = \frac{9.11 \times 10^{-31}}{1.81 \times 10^{29} \times (1.602 \times 10^{-19})^2 \times 2.8 \times 10^{-8}}.$$

Denominator:

$$1.81 \times 10^{29} \times 2.566 \times 10^{-38} = 4.64 \times 10^{-9}, \quad \times 2.8 \times 10^{-8} = 1.30 \times 10^{-16}.$$

Divide:

$$\tau = \frac{9.11 \times 10^{-31}}{1.30 \times 10^{-16}} = \boxed{7.0 \times 10^{-15} \text{ s}}.$$

Mean free path:

$$\ell = v_F \tau = 2.03 \times 10^6 \times 7.0 \times 10^{-15} = \boxed{14 \text{ nm}}.$$

(d) 2D electron gas

$n_{2D} = 3 \times 10^{15} \text{ m}^{-2}$, $\rho_{2D} = 10^3 \Omega$ (per square).

Fermi wavevector ($d = 2$, $a_2 = 2\pi$):

$$k_F = \sqrt{2\pi n_{2D}} = \sqrt{2\pi \times 3 \times 10^{15}}.$$

Argument:

$$2\pi \times 3 \times 10^{15} = 1.885 \times 10^{16}.$$

Square root:

$$k_F = 1.37 \times 10^8 \text{ m}^{-1}.$$

Fermi energy:

$$E_F = \frac{\hbar^2 k_F^2}{2m_e} = \frac{1.113 \times 10^{-68} \times (1.37 \times 10^8)^2}{1.822 \times 10^{-30}}.$$

Square:

$$(1.37 \times 10^8)^2 = 1.876 \times 10^{16}.$$

Numerator:

$$1.113 \times 10^{-68} \times 1.876 \times 10^{16} = 2.09 \times 10^{-52}.$$

Divide:

$$E_F = \frac{2.09 \times 10^{-52}}{1.822 \times 10^{-30}} = 1.15 \times 10^{-22} \text{ J} = \boxed{7.2 \times 10^{-4} \text{ eV}}.$$

(Free electron mass assumed; in a real GaAs 2DEG $m^* \approx 0.067m_e$ would raise this $\sim 15\times$.)

Fermi temperature:

$$T_F = 1.15 \times 10^{-22} / 1.381 \times 10^{-23} = \boxed{8.3 \text{ K}}.$$

Fermi velocity:

$$v_F = \hbar k_F / m_e = 1.055 \times 10^{-34} \times 1.37 \times 10^8 / 9.11 \times 10^{-31} = 1.59 \times 10^4 \text{ m s}^{-1}.$$

Drude in 2D: $\sigma_{2D} = n_{2D}e^2\tau/m_e$, so $\rho_{2D} = m_e/(n_{2D}e^2\tau)$:

$$\tau = \frac{m_e}{n_{2D}e^2\rho_{2D}} = \frac{9.11 \times 10^{-31}}{3 \times 10^{15} \times 2.566 \times 10^{-38} \times 10^3}.$$

Denominator:

$$3 \times 10^{15} \times 2.566 \times 10^{-38} = 7.70 \times 10^{-23}, \times 10^3 = 7.70 \times 10^{-20}.$$

Divide:

$$\tau = \frac{9.11 \times 10^{-31}}{7.70 \times 10^{-20}} = \boxed{1.2 \times 10^{-11} \text{ s}}.$$

Mean free path:

$$\ell = v_F \tau = 1.59 \times 10^4 \times 1.2 \times 10^{-11} = \boxed{0.19 \text{ } \mu\text{m}}.$$

Question 3 — Spin- $\frac{1}{2}$ paramagnet and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Problem. Derive $M = M_s \tanh(\eta B/T)$ and χ . Apply Hund's rules to Cu^{2+} , explain the role of the lattice and contrast with elemental Cu, then evaluate χ and the alignment fraction at 300 K and 2 K for $B = 0.05 \text{ T}$ and 5 T .

(a) Magnetization and susceptibility

Two-level Zeeman system: energies $\mp \frac{1}{2} g \mu_B B$ for $S_z = \pm \frac{1}{2}$ ($g \simeq 2$).

Boltzmann populations:

$$P_{\pm} = \frac{e^{\pm g \mu_B B / (2k_B T)}}{Z}, \quad Z = 2 \cosh\left(\frac{g \mu_B B}{2k_B T}\right).$$

Magnetic moment per spin (along $\mathbf{B}$):

$$\langle \mu_z \rangle = \frac{1}{2} g \mu_B (P_+ - P_-).$$

Substitute the populations:

$$\langle \mu_z \rangle = \frac{1}{2} g \mu_B \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad x = \frac{g \mu_B B}{2k_B T}.$$

Recognise hyperbolic tangent:

$$\langle \mu_z \rangle = \frac{1}{2} g \mu_B \tanh x.$$

Multiply by number density N/V :

$$\boxed{M = M_s \tanh(\eta B/T), \quad M_s = \frac{N g \mu_B}{2V}, \quad \eta = \frac{g \mu_B}{2k_B}.} \quad (7)$$

Small-field susceptibility: linearise $\tanh x \approx x$ for $\eta B/T \ll 1$:

$$M \approx M_s \frac{\eta B}{T} = \frac{N g \mu_B}{2V} \cdot \frac{g \mu_B B}{2k_B T}.$$

Use $\chi = \mu_0 M/B$ (SI volume susceptibility):

$$\boxed{\chi = \frac{\mu_0 N g^2 \mu_B^2}{4V k_B T} = \frac{\mu_0 n}{k_B T} \left(\frac{1}{2} g \mu_B\right)^2.} \quad (8)$$

This is Curie's law $\chi \propto 1/T$.

(b) Cu²⁺ Hund's-rule moment

Cu²⁺ configuration 3d⁹: one hole in the 3d shell.

Hund's rules on the hole: $S = \frac{1}{2}$, $L = 2$, $J = L + S = \frac{5}{2}$ (more-than-half-filled shell).

Free-ion Landé g -factor:

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)} = \frac{3}{2} + \frac{0.75 - 6}{2 \cdot 8.75} = \frac{3}{2} - 0.300 = 1.20.$$

Effective Bohr-magneton number:

$$p = g_J \sqrt{J(J+1)} = 1.20 \sqrt{8.75} = 3.55.$$

$$\mu_{\text{eff}}^{\text{free}} = 3.55 \mu_B \text{ (free ion, full } J\text{).}$$

In the crystal the orbital moment is *quenched* by the cubic/octahedral ligand field: $\langle L_z \rangle = 0$. Only spin contributes, $g \rightarrow 2$, $S = \frac{1}{2}$:

$$p_{\text{spin}} = 2 \sqrt{S(S+1)} = 2 \sqrt{0.75} = 1.73, \mu_{\text{eff}} = 1.73 \mu_B.$$

The SO₄ groups and water of hydration physically separate Cu²⁺ ions, suppressing direct d -orbital overlap and exchange. Cu²⁺ moments are therefore non-interacting paramagnetic centres rather than ordering magnetically.

Pure elemental Cu has configuration 3d¹⁰4s¹: filled d -shell (no net moment) plus a delocalised 4s conduction band. It exhibits weak Pauli paramagnetism (overwhelmed in practice by core diamagnetism), so Cu metal is overall *diamagnetic*.

(c) Susceptibility and alignment fraction

Number density of Cu²⁺ ions:

$$n = \frac{N_A}{V_m} = \frac{6.022 \times 10^{23}}{1.092 \times 10^{-4}} = 5.515 \times 10^{27} \text{ m}^{-3}.$$

Spin-only effective moment squared (for the susceptibility formula):

$$g^2 S(S+1) = 4 \times 0.75 = 3.$$

Generalised Curie formula (replacing $(g\mu_B/2)^2 \rightarrow g^2 S(S+1)\mu_B^2/3$ in (??) for general S):

$$\chi = \frac{\mu_0 n g^2 S(S+1) \mu_B^2}{3k_B T}.$$

Evaluate the constant prefactor:

$$\mu_0 n g^2 S(S+1) \mu_B^2 = 4\pi \times 10^{-7} \times 5.515 \times 10^{27} \times 3 \times (9.274 \times 10^{-24})^2.$$

Compute step by step:

$$(9.274 \times 10^{-24})^2 = 8.602 \times 10^{-47}.$$

$$4\pi \times 10^{-7} \times 5.515 \times 10^{27} = 6.929 \times 10^{21}.$$

$$6.929 \times 10^{21} \times 3 = 2.079 \times 10^{22}.$$

$$2.079 \times 10^{22} \times 8.602 \times 10^{-47} = 1.789 \times 10^{-24}.$$

Divide by $3k_B T$:

$$\chi(T) = \frac{1.789 \times 10^{-24}}{3 \times 1.381 \times 10^{-23} \times T} = \frac{4.32 \times 10^{-2}}{T}.$$

At 300 K:

$$\boxed{\chi(300 \text{ K}) = 1.44 \times 10^{-4}} \text{ (dimensionless, SI).}$$

At 2 K:

$$\boxed{\chi(2 \text{ K}) = 2.16 \times 10^{-2}}.$$

Alignment fraction. For spin- $\frac{1}{2}$ in field B , fraction aligned with $\mathbf{B}$ is $f = \tanh x$ with $x = g\mu_B B / (2k_B T) = \mu_B B / (k_B T)$.

Reference scale:

$$\frac{\mu_B}{k_B} = \frac{9.274 \times 10^{-24}}{1.381 \times 10^{-23}} = 0.6717 \text{ K T}^{-1}.$$

$T = 300 \text{ K}$.

$$B = 0.05 \text{ T} : x = 0.6717 \times 0.05/300 = 1.12 \times 10^{-4}, \tanh x \approx 1.12 \times 10^{-4}.$$

$$\boxed{f \approx 1.1 \times 10^{-4}}.$$

$$B = 5 \text{ T} : x = 0.6717 \times 5/300 = 1.12 \times 10^{-2}, \tanh x \approx 1.12 \times 10^{-2}.$$

$$\boxed{f \approx 1.1 \times 10^{-2}}.$$

$T = 2 \text{ K}$.

$$B = 0.05 \text{ T} : x = 0.6717 \times 0.05/2 = 1.68 \times 10^{-2}, \tanh x \approx 1.68 \times 10^{-2}.$$

$$\boxed{f \approx 1.7 \times 10^{-2}}.$$

$$B = 5 \text{ T} : x = 0.6717 \times 5/2 = 1.68, \tanh(1.68) = 0.933.$$

$$\boxed{f \approx 0.93 \text{ (near saturation)}}.$$

Question 4 — Lattice vibrations of a 1D monatomic chain

Problem. Outline experimental probes of phonons; derive the dispersion of a 1D monatomic chain; sketch $\omega(k)$, $v_g(k)$, $g(\omega)$, and $C(T)$ for $a = 0.2 \text{ nm}$, $v_s = 4 \text{ km s}^{-1}$.

(a) Experimental methods

- **Inelastic neutron scattering.** Thermal neutrons ($\hbar\omega \sim \text{meV}$, $\lambda \sim \text{\AA}$) match phonon energy and momentum scales. Energy and momentum conservation, $E_i - E_f = \hbar\omega(\mathbf{q})$ and $\mathbf{k}_i - \mathbf{k}_f = \mathbf{q} + \mathbf{G}$, map full $\omega(\mathbf{q})$ across the BZ on triple-axis or time-of-flight spectrometers.
- **Inelastic X-ray scattering** (synchrotron, meV resolution): same kinematics, but photon $\omega \gg$ phonon ω , so the energy transfer is a tiny fraction of the incident energy; useful for small samples (high-pressure cells).
- **Raman scattering** (visible photons): probes optic phonons at $\mathbf{q} \approx 0$ (Stokes/anti-Stokes shift); selection rules require Raman-active modes.

- **Infrared absorption/reflectivity:** dipole-active optical phonons ($q \approx 0$); reststrahlen band identifies ω_{TO} , ω_{LO} .
- **Heat capacity vs T** (Debye fits): integrated information on $g(\omega)$.

(b) Equation of motion and dispersion

Atom n at equilibrium $x_n^0 = na$, displacement $u_n(t)$. Spring force from neighbour $n + 1$ proportional to extension $u_{n+1} - u_n$; from $n - 1$ proportional to $u_{n-1} - u_n$.

Newton's second law:

$$m\ddot{u}_n = C(u_{n+1} - u_n) + C(u_{n-1} - u_n).$$

Group:

$$m\ddot{u}_n = C(u_{n+1} - 2u_n + u_{n-1}).$$

Plane-wave ansatz:

$$u_n(t) = u_0 e^{i(kna - \omega t)}.$$

Substitute:

$$-m\omega^2 = C(e^{ika} - 2 + e^{-ika}).$$

Use $e^{ika} + e^{-ika} = 2 \cos ka$:

$$-m\omega^2 = 2C(\cos ka - 1).$$

Apply $1 - \cos ka = 2 \sin^2(ka/2)$:

$$m\omega^2 = 4C \sin^2(ka/2).$$

Take positive root:

$$\boxed{\omega(k) = 2\sqrt{C/m} |\sin(ka/2)|.} \quad (9)$$

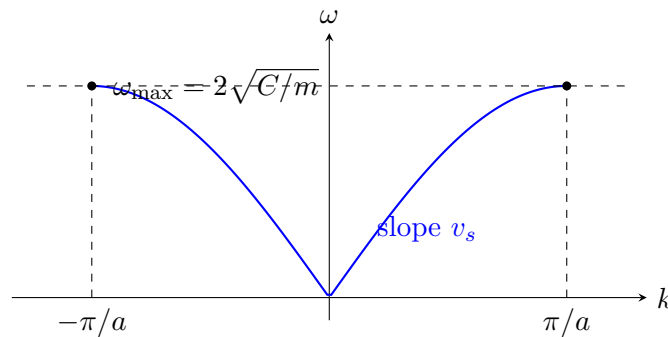
(c) Sketches

Long-wavelength sound speed: $\omega = v_s |k|$ for $|k|a \ll 1$, giving $v_s = a\sqrt{C/m}$. Maximum frequency:

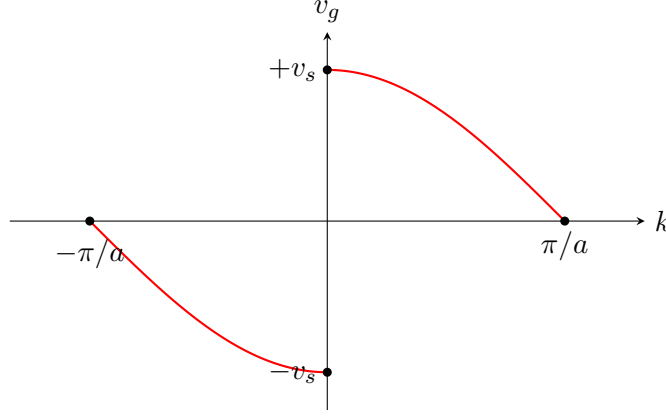
$$\omega_{\max} = 2\sqrt{C/m} = 2v_s/a = 2 \times 4 \times 10^3 / 0.2 \times 10^{-9} = 4 \times 10^{13} \text{ rad s}^{-1}.$$

First Brillouin zone: $k \in [-\pi/a, \pi/a]$, with $\pi/a = 1.57 \times 10^{10} \text{ m}^{-1}$.

(i) Dispersion $\omega(k)$.



(ii) **Group velocity** $v_g(k) = d\omega/dk = a\sqrt{C/m} \cos(ka/2) \operatorname{sgn}(\sin(ka/2))$ on $|k| < \pi/a$. For $0 < k < \pi/a$: $v_g = a\sqrt{C/m} \cos(ka/2) = v_s \cos(ka/2)$, dropping from v_s at $k = 0$ to 0 at $k = \pi/a$ (and odd in k).



(Group velocity is discontinuous at $k = 0$ if absolute value is taken in (??); physically, v_g changes sign through $k = 0$ as drawn.)

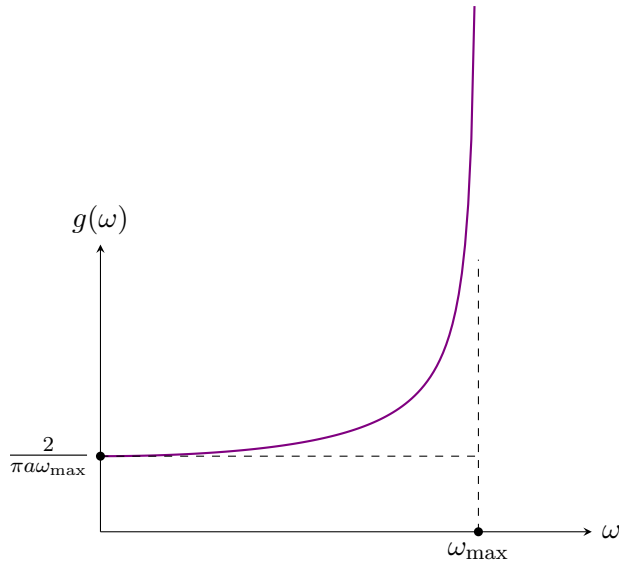
(iii) **Density of states** $g(\omega)$. 1D, both $\pm k$ states:

$$g(\omega) = \frac{2}{2\pi} \frac{1}{|d\omega/dk|} = \frac{1}{\pi} \frac{1}{a\sqrt{C/m} |\cos(ka/2)|}.$$

Use $\cos(ka/2) = \sqrt{1 - \sin^2(ka/2)} = \sqrt{1 - (\omega/\omega_{\max})^2}$:

$$g(\omega) = \frac{2}{\pi a \omega_{\max}} \frac{1}{\sqrt{1 - (\omega/\omega_{\max})^2}}, \quad 0 < \omega < \omega_{\max}.$$

Constant low- ω value $2/(\pi a \omega_{\max})$, van Hove divergence at $\omega_{\max}$.



(iv) **Heat capacity** $C_V(T)$. Low T ($k_B T \ll \hbar \omega_{\max}$): linear-dispersion phonons give 1D Debye behaviour $C_V \propto T$ (one polarization). High T ($k_B T \gg \hbar \omega_{\max}$): equipartition $C_V \rightarrow Nk_B$ per atom (Dulong–Petit limit, classical).

Crossover scale: $\theta_D = \hbar \omega_{\max}/k_B$.

$$\theta_D = \hbar \omega_{\max}/k_B = (1.055 \times 10^{-34} \times 4 \times 10^{13})/1.381 \times 10^{-23} = 306 \text{ K}.$$

